The Harmonic Transform

A Recurrence-Defined Polynomial Basis

Carolina Johnson (CJ)

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Abstract

This paper does not propose a faster transform and does not compete with the FFT or with Chebyshev DCT methods. It characterizes a recurrence-defined family of polynomial bases on $[-1, 1]$, the recurrence basis, and the Harmonic Transform (HT) that is its coordinate map, and it states when the family is the appropriate tool. The basis functions are generated by $H_0 = 0$, $H_1 = x$, $H_{k+1} = 2xH_k - \Phi_k H_{k-1}$ with an arbitrary real sequence Φ_k, and the sequence selects a basis of a fixed space: for every Φ, $\mathrm{span}\{H_1, \dots, H_K\}$ is the space V_K of polynomials of degree at most K that vanish at 0, and two sequences are related by a unit lower triangular change of basis. On any grid where the basis matrix has full column rank, the least-squares reconstruction is the orthogonal projection onto the same subspace for every Φ, so the reconstruction does not depend on Φ while the coefficients do. The Harmonic Transform (HT) is the coordinate map of the recurrence basis. Its extension to sampled data is the standard least-squares fit in that basis, and the whole basis matrix is generated by the recurrence itself, with no expanded coefficients. The least-squares fit is leakage-free by construction: it imposes no periodic extension, the residual is orthogonal to the fitted space, and a signal in V_K is reconstructed exactly with zero coordinates on every added basis function, provided the basis matrix, including the added basis functions, has full column rank. For $\Phi_k = 1$ the basis is $H_k = xU_{k-1}(x)$ with U_{k-1} the Chebyshev polynomial of the second kind, and $H_k(\cos\theta) = \cos\theta \sin(k\theta)/\sin\theta$. For every sequence $\Phi_k > 0$ the recurrence basis is orthogonal for an explicit inner product determined by the measure of Favard's theorem, and the Stieltjes procedure applied to $x^2\sigma$ returns the recurrence coefficients whose basis is orthogonal for a given symmetric weight σ; the Jacobi matrix of the coefficients gives the Gauss rule of that measure. The paper gives an explicit expansion of H_k in the entries of Φ, the exact rank condition on the nodes, operation counts, a table of when to use this basis and when to use Fourier or Chebyshev methods, and numerical results as exact rational values, high-precision values, and binary64 observations with the computing environment stated.

1 Definition

Throughout, $\mathcal{P}_K$ denotes the space of real polynomials of degree at most K, $\|\cdot\|_2$ the Euclidean norm, and $\varepsilon = 2^{-52}$ the machine epsilon of IEEE 754 binary64 arithmetic.

Definition 1.1 (Recurrence basis)**.** Let $(\Phi_k)_{k\geq 1}$ be a sequence of real numbers, called the recurrence coefficients. The basis functions H_k associated with Φ, called the *recurrence basis*, are defined by

$$H_0(x) = 0, \qquad H_1(x) = x, \qquad H_{k+1}(x) = 2x\,H_k(x) - \Phi_k\,H_{k-1}(x) \quad (k \geq 1).$$

Remark 1.2. The coefficient Φ_1 multiplies $H_0 = 0$, so Φ_1 does not enter any H_k. Every result below uses no property of Φ_k beyond being a real number, with exceptions that impose a condition on the coefficients: Proposition 4.1 uses $\Phi_0 > 0$, and Remark 2.4 and Section 5 use $\Phi_k > 0$ for all $k \geq 2$.

The first members follow by direct substitution:

$$
\begin{aligned}
H_2 &= 2x \cdot x - \Phi_1 \cdot 0 = 2x^2, \\
H_3 &= 2x\,(2x^2) - \Phi_2\, x = 4x^3 - \Phi_2 x, \\
H_4 &= 2x\,(4x^3 - \Phi_2 x) - \Phi_3\,(2x^2) = 8x^4 - 2\Phi_2 x^2 - 2\Phi_3 x^2 = 8x^4 - 2(\Phi_2 + \Phi_3)x^2, \\
H_5 &= 2x\,(8x^4 - 2(\Phi_2 + \Phi_3)x^2) - \Phi_4\,(4x^3 - \Phi_2 x) \\
&\quad = 16x^5 - 4(\Phi_2 + \Phi_3)x^3 - 4\Phi_4 x^3 + \Phi_2\Phi_4 x = 16x^5 - 4(\Phi_2 + \Phi_3 + \Phi_4)x^3 + \Phi_2\Phi_4\, x.
\end{aligned}
$$

Definition 1.3. Two sequences are used by name. The *constant sequence* is $\Phi_k = \Phi_0$ with $\Phi_0 > 0$. The *geometric sequence* is $\Phi_k = \Phi_0 \lambda^{-k}$ with $\Phi_0 > 0$ and $\lambda > 0$; it is the constant sequence when $\lambda = 1$.

Purpose and scope. This paper does not propose a faster transform and does not compete with the FFT or with Chebyshev methods. It records a recurrence-defined polynomial basis, its span, its coordinates, its conditioning, and the least-squares projection it induces. In this paper the Harmonic Transform (HT) denotes the coordinate map of the recurrence basis (Definition 3.2), and the word harmonic is used in the sense of Remark 4.3; the name is used in the sense defined here. Section 13 states when this basis is the appropriate tool and when a Fourier or Chebyshev method is.

Organization and status of the results. Sections 2 to 4 study the recurrence basis and its coordinate map, the HT. Section 5 connects the recurrence coefficients to orthogonal polynomials, weights, and Gauss quadrature. Section 6 defines the least-squares fit on a grid, Section 7 states exact reconstruction and the sense in which the fit is leakage-free, and Section 8 describes the evaluation of the basis matrix by the recurrence. Each result is labeled. A *Proposition* is proved here from the recurrence by induction, dimension counting, root counting, or direct computation. A *Standard fact* is a result of linear algebra or numerical analysis that is stated with a reference, and is proved here only where the proof is short. The least-squares solution, the pseudoinverse, the orthogonal projector, exact reconstruction on the column space, and the singular value decomposition are standard; none is claimed as new. The contribution of this paper is the definition of the family H_k and the identification of its span, coordinates, and conditioning. Section 12 reports computations; it separates exact rational arithmetic, high-precision arithmetic, and binary64 arithmetic, and states which values depend on the computing environment.

2 Structure of the basis functions

Proposition 2.1 (Divisibility, degree, leading coefficient)**.** *For every real sequence Φ define polynomials $q_0 = 1$, $q_1 = 2x$, and $q_k = 2x\, q_{k-1} - \Phi_k\, q_{k-2}$ for $k \geq 2$. Then*

(a) $H_k = x\, q_{k-1}$ for every $k \geq 1$;
(b) q_k has degree k and leading coefficient 2^k for every $k \geq 0$.

Consequently $H_k(0) = 0$, $\deg H_k = k$, and the leading coefficient of H_k is 2^{k-1}.

Proof. (a) Induction on k. For $k = 1$, $H_1 = x = x\, q_0$. For $k = 2$, $H_2 = 2x^2 = x\, q_1$. Let $k \geq 2$ and assume $H_j = x\, q_{j-1}$ for $j \leq k$. Then

$$
H_{k+1} = 2x\, H_k - \Phi_k H_{k-1} = 2x \cdot x\, q_{k-1} - \Phi_k\, x\, q_{k-2} = x\,(2x\, q_{k-1} - \Phi_k q_{k-2}) = x\, q_k.
$$

(b) Induction on k. $q_0 = 1$ has degree 0 and leading coefficient 2^0; $q_1 = 2x$ has degree 1 and leading coefficient 2^1. Let $k \geq 2$. The term $2x\, q_{k-1}$ has degree k and leading coefficient $2 \cdot 2^{k-1} = 2^k$, and the term $\Phi_k q_{k-2}$ has degree at most $k-2 < k$. Hence q_k has degree k and leading coefficient 2^k.

The consequences follow from (a) and (b): $H_k = x\, q_{k-1}$ vanishes at 0, has degree $1 + (k-1) = k$, and has leading coefficient 2^{k-1}. □

For integers $m \geq 0$ and $j \geq 0$ let $E_j(m)$ denote the sum of $\prod_{i \in S} \Phi_i$ over all subsets $S \subseteq \{2, \dots, m\}$ with $|S| = j$ that contain no two consecutive integers. The empty product is 1, so $E_0(m) = 1$ for all $m \geq 0$, and $E_j(m) = 0$ when no such subset exists. Set $E_{-1}(m) = 0$.

Proposition 2.2 (Explicit expansion)**.** *For every real sequence Φ and every $k \geq 1$,*

$$H_k(x) = \sum_{j \geq 0} (-1)^j\, 2^{\,k-1-2j}\, E_j(k-1)\, x^{\,k-2j},$$

where the terms with $E_j(k-1) = 0$ vanish, and the remaining terms have $k - 2j \geq 1$.

Proof. By Proposition 2.1(a) it suffices to prove, for all $k \geq 0$,

$$q_k(x) = \sum_{j \geq 0} (-1)^j\, 2^{\,k-2j}\, E_j(k)\, x^{\,k-2j}, \tag{1}$$

because $H_k = x\, q_{k-1}$ then gives the stated formula. For $k = 0$ the right side is $E_0(0) = 1 = q_0$. For $k = 1$ it is $2x\, E_0(1) = 2x = q_1$, since $E_j(1) = 0$ for $j \geq 1$ (the set $\{2, \dots, 1\}$ is empty). Let $k \geq 2$ and assume (1) for $k-1$ and $k-2$. Compare coefficients of x^{k-2j} in $q_k = 2x\, q_{k-1} - \Phi_k q_{k-2}$.

The term $2x\, q_{k-1}$ contributes $2 \cdot (-1)^j 2^{\,k-1-2j} E_j(k-1) = (-1)^j 2^{\,k-2j} E_j(k-1)$.

For the term $-\Phi_k q_{k-2}$, write $x^{k-2j} = x^{(k-2)-2(j-1)}$. The coefficient of this power in q_{k-2} is $(-1)^{j-1} 2^{\,k-2j} E_{j-1}(k-2)$, so the term contributes $(-1)^j 2^{\,k-2j}\, \Phi_k\, E_{j-1}(k-2)$.

The sum of the two contributions is $(-1)^j 2^{\,k-2j} [E_j(k-1) + \Phi_k E_{j-1}(k-2)]$. The subsets $S \subseteq \{2, \dots, k\}$ with $|S| = j$ and no two consecutive integers split into those not containing k, which are counted by $E_j(k-1)$, and those containing k; the latter cannot contain $k-1$, so they consist of k together with a non-consecutive subset of $\{2, \dots, k-2\}$ of size $j-1$, and contribute $\Phi_k E_{j-1}(k-2)$. Hence the bracket equals $E_j(k)$, which proves (1). □

Remark 2.3. For $k = 6$ the subsets of $\{2, 3, 4, 5\}$ without consecutive integers have sizes $0, 1, 2$, with $E_1(5) = \Phi_2 + \Phi_3 + \Phi_4 + \Phi_5$ and $E_2(5) = \Phi_2\Phi_4 + \Phi_2\Phi_5 + \Phi_3\Phi_5$. Proposition 2.2 gives

$$H_6 = 32x^6 - 8(\Phi_2 + \Phi_3 + \Phi_4 + \Phi_5)\, x^4 + 2(\Phi_2\Phi_4 + \Phi_2\Phi_5 + \Phi_3\Phi_5)\, x^2.$$

The expansions of H_3, H_4, H_5 in Section 1 agree with the proposition.

Remark 2.4 (Relation to orthogonal polynomials). Let $p_k = q_k / 2^k$. Then $p_0 = 1$, $p_1 = x$, and $p_k = x\, p_{k-1} - (\Phi_k/4)\, p_{k-2}$ for $k \geq 2$, a symmetric three-term recurrence. If $\Phi_k > 0$ for all $k \geq 2$, Favard's theorem states that the p_k are the monic orthogonal polynomials of a positive measure on $\mathbb{R}$ [5, 4]; for $\Phi_k = 1$ they are $U_k/2^k$. Section 5 states the orthogonality of the functions $H_k = x\, q_{k-1}$ themselves. No result in Sections 3 to 4 or in Sections 6 to 13 uses orthogonality.

3 Span, coordinates, and change of basis

Proposition 3.1 (The span is independent of Φ)**.** *For every real sequence Φ and every $K \geq 1$,*

$$V_K := \operatorname{span}\{H_1, \dots, H_K\} = \{p \in \mathcal{P}_K : \ p(0) = 0\}.$$

Proof. Let $W_K = \{p \in \mathcal{P}_K : p(0) = 0\}$. By Proposition 2.1, each H_k with $k \leq K$ belongs to W_K, and $H_1, \dots, H_K$ have the distinct degrees $1, \dots, K$, so they are linearly independent and $\dim V_K = K$. The map $p \mapsto p(0)$ is a nonzero linear functional on $\mathcal{P}_K$, which has dimension $K+1$, so its kernel W_K has dimension K. Since $V_K \subseteq W_K$ and both have dimension K, $V_K = W_K$. □

Definition 3.2 (Harmonic Transform of order K)**.** The *synthesis map* is $S_K : \mathbb{R}^K \to V_K$, $S_K(a) = \sum_{k=1}^{K} a_k H_k$. By linear independence and Proposition 3.1, S_K is a linear bijection. The *Harmonic Transform* (HT) of order K of $p \in V_K$ is the coefficient vector $a = S_K^{-1}(p)$, that is, the coordinate vector of p in the basis $H_1, \dots, H_K$.

The HT is therefore the coordinate map of a basis, a standard construction; its content is in the basis. Section 6 extends the coordinate map to sampled data that do not come from an element of V_K.

Proposition 3.3 (Unit lower triangular change of basis)**.** *Let Φ and Φ' be real sequences with basis functions H_k and H'_k. For every K there is a unique matrix $T \in \mathbb{R}^{K \times K}$, unit lower triangular, with*

$$H'_k = \sum_{j=1}^{k} T_{kj} H_j \qquad (1 \leq k \leq K).$$

Proof. For $k = 1$, $H'_1 = H_1 = x$. For $k \geq 2$ let $D = H'_k - H_k = x\,(q'_{k-1} - q_{k-1})$ (Proposition 2.1(a)). Both q'_{k-1} and q_{k-1} have degree $k-1$ and leading coefficient 2^{k-1} (Proposition 2.1(b)), so $q'_{k-1} - q_{k-1}$ has degree at most $k-2$, and D has degree at most $k-1$ and satisfies $D(0) = 0$. By Proposition 3.1, $D \in V_{k-1} = \operatorname{span}\{H_1, \dots, H_{k-1}\}$. Hence $H'_k = H_k + \sum_{j<k} T_{kj} H_j$, so $T_{kk} = 1$. Uniqueness follows from the linear independence of $H_1, \dots, H_K$. □

Corollary 3.4 (Coordinates depend on Φ; the element does not)**.** *Let $p \in V_K$ with $p = \sum_k a_k H_k = \sum_k a'_k H'_k$. Then $a = T^\top a'$, and $\det T = 1$.*

Proof. $\sum_k a'_k H'_k = \sum_k a'_k \sum_j T_{kj} H_j = \sum_j (\sum_k T_{kj} a'_k) H_j = \sum_j (T^\top a')_j H_j$. The coordinates with respect to the independent set $H_1, \dots, H_K$ are unique, so $a = T^\top a'$. The determinant of a unit triangular matrix is 1. □

4 The Chebyshev connection and the angle form

The Chebyshev polynomials of the second kind are defined by $U_0(t) = 1$, $U_1(t) = 2t$, and $U_m(t) = 2t\,U_{m-1}(t) - U_{m-2}(t)$ for $m \geq 2$.

Proposition 4.1 (Constant sequence)**.** *If $\Phi_k = \Phi_0 > 0$ for all k, then for every $k \geq 1$*

$$H_k(x) = \Phi_0^{(k-1)/2}\; x\; U_{k-1}(x/\sqrt{\Phi_0}).$$

In particular, for $\Phi_k = 1$, $H_k(x) = x\,U_{k-1}(x)$.

Proof. Define $u_k(x) = \Phi_0^{k/2} U_k(x/\sqrt{\Phi_0})$. Then $u_0 = 1$ and $u_1 = \Phi_0^{1/2} \cdot 2x/\sqrt{\Phi_0} = 2x$. For $k \geq 2$,

$$u_k = \Phi_0^{k/2}\Big[\tfrac{2x}{\sqrt{\Phi_0}} U_{k-1} - U_{k-2}\Big] = 2x\,\Phi_0^{(k-1)/2} U_{k-1} - \Phi_0\,\Phi_0^{(k-2)/2} U_{k-2} = 2x\,u_{k-1} - \Phi_0\,u_{k-2}.$$

This is the recurrence of q_k in Proposition 2.1 with $\Phi_k = \Phi_0$, with the same initial values, so $q_k = u_k$ for all k. By Proposition 2.1(a), $H_k = x\,q_{k-1} = x\,u_{k-1}$, which is the stated formula. For $\Phi_0 = 1$, $u_k = U_k$. □

Proposition 4.2 (Angle form for $\Phi_k = 1$)**.** *For $\Phi_k = 1$ and $\theta \in (0, \pi)$,*

$$H_k(\cos\theta) = \cos\theta\,\frac{\sin(k\theta)}{\sin\theta},$$

and $H_k(1) = k$, $H_k(-1) = (-1)^k k$.

Proof. Let $w_m = \sin((m+1)\theta)/\sin\theta$. Then $w_0 = 1$ and $w_1 = \sin 2\theta/\sin\theta = 2\cos\theta$. The identity $\sin(A+B) + \sin(A-B) = 2\sin A\cos B$ with $A = m\theta$, $B = \theta$ gives $\sin((m+1)\theta) + \sin((m-1)\theta) = 2\sin(m\theta)\cos\theta$; dividing by $\sin\theta$ gives $w_m + w_{m-2} = 2\cos\theta\,w_{m-1}$. So w_m satisfies the recurrence of $U_m(\cos\theta)$ with the same initial values, and $U_m(\cos\theta) = \sin((m+1)\theta)/\sin\theta$. By Proposition 4.1, $H_k(\cos\theta) = \cos\theta\,U_{k-1}(\cos\theta) = \cos\theta\,\sin(k\theta)/\sin\theta$.

At the endpoints, $U_0(1) = 1$, $U_1(1) = 2$, and if $U_j(1) = j+1$ for $j < m$ then $U_m(1) = 2m - (m-1) = m+1$; hence $U_{k-1}(1) = k$ and $H_k(1) = 1 \cdot k$. Likewise $U_0(-1) = 1$, $U_1(-1) = -2$, and if $U_j(-1) = (-1)^j(j+1)$ for $j < m$ then $U_m(-1) = -2(-1)^{m-1}m - (-1)^{m-2}(m-1) = (-1)^m(2m - (m-1)) = (-1)^m(m+1)$; hence $H_k(-1) = (-1)\cdot(-1)^{k-1}k = (-1)^k k$. □

Remark 4.3 (Use of the word harmonic)*.* In this paper *harmonic* refers to the integer-multiple structure of Proposition 4.2: for $\Phi_k = 1$ the factor $\sin(k\theta)$ in H_k has angular frequency k, the k-th integer multiple of the fundamental angular frequency 1. For other sequences Φ the functions H_k are a different basis of the same space V_K (Proposition 3.1), and no statement about their form in θ is made. The name Harmonic Transform is used for the coordinate map of Definition 3.2 and in no other sense.

5 Orthogonality and the Jacobi matrix

This section connects the recurrence coefficients to standard orthogonal polynomial theory [5, 4]. It records which inner product the recurrence basis is orthogonal for, how to compute Φ from a given weight, and how Φ gives a Gauss quadrature rule. The results are standard and are stated in the form in which they are used. A *measure* here is a positive Borel measure on $\mathbb{R}$ with finite moments and infinite support, and a measure is *symmetric* if it is invariant under $x \mapsto -x$.

Standard fact 5.1 (Favard's theorem and symmetric measures)**.** (a) If $\Phi_k > 0$ for all $k \geq 2$, there is a symmetric measure ν for which $p_k = q_k/2^k$ are the monic orthogonal polynomials. If (Φ_k) is bounded, ν has bounded support, and a measure with bounded support is determined uniquely by its moments, so ν is unique.

(b) Conversely, let ν be a symmetric measure with monic orthogonal polynomials p_k and $h_k = \int p_k^2\,d\nu$. Then $p_0 = 1$, $p_1 = x$, and $p_{k+1} = x\,p_k - b_k\,p_{k-1}$ for $k \geq 1$, with $b_k = h_k/h_{k-1} > 0$.

For (b): $p_{k+1} - x\,p_k$ has degree at most k and $\int (p_{k+1} - x\,p_k)\,p_j\,d\nu = -\int x\,p_k p_j\,d\nu$ for $j \le k$; this vanishes for $j < k-1$ because $x\,p_j$ has degree less than k, for $j = k$ because $p_k^2 x$ is odd (the p_k have the parity of k and ν is symmetric), and for $j = k-1$ it equals $-\int p_k\,(x\,p_{k-1})\,d\nu = -h_k$. Dividing by h_{k-1} gives the coefficient $-b_k$. Part (a) is Favard's theorem [5, 4]. Comparing (b) with $p_k = x\,p_{k-1} - (\Phi_k/4)\,p_{k-2}$ gives

$$\Phi_k = 4\,b_{k-1} \qquad (k \ge 2). \tag{2}$$

The coefficient Φ_1 is not used (Remark 1.2), so the first coefficient is $b_1 = \Phi_2/4$, and the first off-diagonal entry of the Jacobi matrix in Standard fact 5.3 is $\sqrt{b_1} = \frac{1}{2}\sqrt{\Phi_2}$. The recurrence of Definition 1.1 has no shift term, so it produces the orthogonal polynomials of symmetric measures only.

Proposition 5.2 (Orthogonality of the recurrence basis)**.** *Let $\Phi_k > 0$ for all $k \ge 2$ and let ν be a measure as in Standard fact 5.1(a). On the space $V = \bigcup_K V_K$ of polynomials vanishing at 0 define*

$$\langle f, g\rangle_\nu = \int \frac{f(x)}{x}\,\frac{g(x)}{x}\,d\nu(x).$$

This is an inner product on V, and the recurrence basis is orthogonal for it:

$$\langle H_j, H_k\rangle_\nu = 0 \ (j \ne k), \qquad \langle H_k, H_k\rangle_\nu = 4^{\,k-1} h_{k-1}.$$

If $\nu = x^2\sigma$ for a measure σ on $[-1,1]$, then $\langle f, g\rangle_\nu = \int fg\,d\sigma$, so the H_k are orthogonal for σ in the usual sense.

Proof. For $f \in V$, $f(x)/x$ is a polynomial, so the integral is finite. It is bilinear and symmetric. If $f \ne 0$ then f/x is a nonzero polynomial, and $\int (f/x)^2\,d\nu > 0$ because ν has infinite support. By Proposition 2.1(a), $H_k/x = q_{k-1} = 2^{k-1} p_{k-1}$, hence

$$\langle H_j, H_k\rangle_\nu = 2^{\,j+k-2}\int p_{j-1}\,p_{k-1}\,d\nu = 2^{\,j+k-2}\,\delta_{jk}\,h_{k-1},$$

which is $4^{k-1}h_{k-1}$ for $j = k$. If $\nu = x^2\sigma$, then $(f/x)(g/x)\,d\nu = fg\,d\sigma$. □

For $\Phi_k = 1$ the measure is $\nu = \frac{2}{\pi}\sqrt{1-x^2}\,dx$, and $\langle H_j, H_k\rangle_\nu = \frac{2}{\pi}\int x^{-2} H_j H_k \sqrt{1-x^2}\,dx$ is finite because $H_j H_k$ vanishes to second order at 0; the corresponding $\sigma = \nu/x^2$ is not a finite measure.

From a weight to the recurrence coefficients. Let σ be a symmetric measure on $[-1,1]$ and $\nu = x^2\sigma$, a symmetric measure with infinite support. By Proposition 5.2, the coefficients $\Phi_k = 4b_{k-1}$ of the monic orthogonal polynomials of ν give a recurrence basis orthogonal for σ. The coefficients are computed by the discretized Stieltjes procedure [5]. Let $(t_i, w_i)_{i=1}^M$ be a quadrature rule for ν, for example $w_i = \omega_i t_i^2 \sigma(t_i)$ for a rule (t_i, ω_i) for the Lebesgue measure.

(S1) Set $p_{-1}(t_i) = 0$, $p_0(t_i) = 1$.
(S2) For $k = 0, 1, \dots, K$: compute $h_k = \sum_i w_i\,p_k(t_i)^2$ and $a_k = \sum_i w_i\,t_i\,p_k(t_i)^2/h_k$; for $k \ge 1$ set $b_k = h_k/h_{k-1}$ (and $b_0 = 0$); set $p_{k+1}(t_i) = (t_i - a_k)\,p_k(t_i) - b_k\,p_{k-1}(t_i)$.
(S3) Return $\Phi_k = 4b_{k-1}$ for $k = 2, \dots, K+1$, and a_k, which vanish for symmetric rules.

The rule must integrate the products $p_k p_j$ with respect to ν accurately for $j, k \le K$; this is checked in Section 12 by repeating the computation with a second rule.

Standard fact 5.3 (Jacobi matrix and Gauss quadrature)**.** Let $\Phi_k > 0$ and let J_K be the symmetric tridiagonal $K \times K$ matrix with zero diagonal and entries $(J_K)_{i,i+1} = (J_K)_{i+1,i} = \sqrt{b_i} = \frac{1}{2}\sqrt{\Phi_{i+1}}$

for $i = 1, \dots, K-1$. The eigenvalues $\tau_1, \dots, \tau_K$ of J_K are the nodes of the K-point Gauss rule for ν, and the weights are $\mu_0\, v_{1,i}^2$ with $\mu_0 = \nu(\mathbb{R})$ and $v_{\cdot,i}$ the normalized eigenvector for τ_i [6]. The rule integrates every polynomial of degree at most $2K-1$ exactly with respect to ν.

Three families of coefficients are known in closed form. For $\Phi_k = 1$, $\nu = \frac{2}{\pi}\sqrt{1-x^2}\,dx$ and $p_k = U_k/2^k$. For $\nu = \frac{1}{2}\,dx$ on $[-1,1]$ the monic Legendre recurrence has $b_k = k^2/(4k^2-1)$, so $\Phi_k = 4(k-1)^2/(4(k-1)^2-1)$, with $\Phi_2 = 4/3$, $\Phi_3 = 16/15$, $\Phi_4 = 36/35$, $\Phi_5 = 64/63$. For $\sigma_1 = 1$ on $[-1,1]$, that is $\nu = x^2\,dx$, the coefficients are rational and are listed in Table 6; the recurrence basis is then orthogonal for the Lebesgue measure on $[-1,1]$. A measure that is not symmetric has $a_k \neq 0$ for some k, and its orthogonal polynomials are outside the recurrence of Definition 1.1; Section 12 reports the shifts for $\sigma = e^x$. The least-squares fit of Section 6 does not use orthogonality and applies to such data unchanged.

6 Least-squares fit on a grid

Definition 6.1 (Basis matrix)**.** For nodes $x_1, \dots, x_N \in [-1,1]$ and $K \geq 1$, the basis matrix $B \in \mathbb{R}^{N\times K}$ has entries $B_{nk} = H_k(x_n)$ for $1 \leq k \leq K$.

The column index starts at $k = 1$ because $H_0 = 0$ would contribute a zero column. For $c \in \mathbb{R}^K$ let $p_c = S_K(c) = \sum_k c_k H_k$; then $(Bc)_n = p_c(x_n)$, so Bc is the vector of samples of $S_K(c)$ and

$$\operatorname{col}(B) = \{(p(x_n))_{n=1}^N : \ p \in V_K\}. \tag{3}$$

Proposition 6.2 (Full column rank)**.** *Let m be the number of distinct nonzero values among $x_1, \dots, x_N$. Then* $\operatorname{rank} B = K$ *if and only if $m \geq K$.*

Proof. Since $H_1, \dots, H_K$ are linearly independent, $p_c = 0$ only for $c = 0$, and $Bc = 0$ if and only if p_c vanishes at every node.

($m \geq K \Rightarrow \operatorname{rank} B = K$.) Suppose $Bc = 0$. The polynomial $p_c \in \mathcal{P}_K$ vanishes at 0 (Proposition 2.1) and at the $m \geq K$ distinct nonzero nodes, so it has at least $K+1$ distinct roots. A nonzero polynomial of degree at most K has at most K roots, so $p_c = 0$ and $c = 0$.

($\operatorname{rank} B = K \Rightarrow m \geq K$.) Suppose $m < K$ and let $z_1, \dots, z_m$ be the distinct nonzero nodes. The polynomial $p(x) = x\prod_{i=1}^m (x - z_i)$ is nonzero, has degree $m+1 \leq K$, and satisfies $p(0) = 0$, so $p \in V_K$ by Proposition 3.1, and $p = p_c$ with $c \neq 0$. Every node is 0 or some z_i, so p_c vanishes at every node and $Bc = 0$. Hence $\operatorname{rank} B < K$. □

Corollary 6.3 (Chebyshev nodes)**.** *The Chebyshev nodes $x_j = \cos((2j-1)\pi/(2N))$, $j = 1, \dots, N$, are the zeros of the Chebyshev polynomial T_N of the first kind. For these nodes B has full column rank K if and only if $K \leq N$ when N is even and $K \leq N-1$ when N is odd.*

Proof. The angles $(2j-1)\pi/(2N)$ lie in $(0,\pi)$ and increase strictly with j; cos is strictly decreasing on $(0,\pi)$, so the nodes are distinct and lie in $(-1,1)$. A node equals 0 if and only if $(2j-1)/(2N) = 1/2$, that is $2j-1 = N$, which has a solution $j \in \{1, \dots, N\}$ if and only if N is odd (then $j = (N+1)/2$). So $m = N$ for even N and $m = N-1$ for odd N, and Proposition 6.2 applies. □

Standard fact 6.4 (Least squares on a grid)**.** Let $B \in \mathbb{R}^{N\times K}$ have rank K and let $f \in \mathbb{R}^N$. Then $B^\top B$ is symmetric positive definite, and $D := (B^\top B)^{-1}B^\top$ satisfies $DB = I_K$. The matrix $P := BD$ is symmetric with $P^2 = P$, $PB = B$, and $B^\top(f - Pf) = 0$. For every $c \in \mathbb{R}^K$,

$$\|Bc - f\|_2^2 = \|B(c - Df)\|_2^2 + \|f - Pf\|_2^2,$$

so $c = Df$ is the unique minimizer of $\|Bc - f\|_2$, and Pf is the orthogonal projection of f onto $\mathrm{col}(B)$.

Proof. For $c \neq 0$, $c^\top B^\top Bc = \|Bc\|_2^2 > 0$ because B has rank K, so $B^\top B$ is positive definite. Next,

$$\begin{aligned} DB &= (B^\top B)^{-1} B^\top B = I_K, \\ P^\top &= B\left((B^\top B)^{-1}\right)^\top B^\top = P, \\ P^2 &= B(B^\top B)^{-1}(B^\top B)(B^\top B)^{-1} B^\top = P, \\ PB &= BDB = B, \\ B^\top(f - Pf) &= B^\top f - B^\top B(B^\top B)^{-1} B^\top f = 0, \end{aligned}$$

where the second line uses that $(B^\top B)^{-1}$ is symmetric. For any c, $Bc - f = B(c - Df) - (f - Pf)$ because $BDf = Pf$. The vector $B(c - Df)$ lies in $\mathrm{col}(B)$ and $f - Pf$ is orthogonal to $\mathrm{col}(B)$, so the squared norm of the sum is the sum of the squared norms, as stated. It is minimal if and only if $B(c - Df) = 0$, that is $c = Df$, since B is injective. □

This is standard numerical linear algebra [3, 2]. The matrix D is the Moore-Penrose pseudoinverse of B when B has full column rank.

Definition 6.5 (Least-squares fit of order K)**.** Let B be the basis matrix of Definition 6.1 with $\operatorname{rank} B = K$. The *least-squares fit of order K* on the nodes $x_1, \dots, x_N$ is the linear map $\mathbb{R}^N \to \mathbb{R}^K$, $f \mapsto c = Df$ with $D = (B^\top B)^{-1} B^\top$. The reconstruction of f is $\hat{f} = Bc = Pf$. For the samples of an element $p \in V_K$ the map returns the Harmonic Transform of p (Standard fact 7.2), so the least-squares fit extends the HT to arbitrary sampled data.

The synthesis equation of the fit is $\hat{f} = Bc$, that is $\hat{f}_n = \sum_k c_k H_k(x_n)$, and the analysis equation is $c = (B^\top B)^{-1} B^\top f$.

Corollary 6.6 (Dependence on Φ)**.** *Let B and B' be the basis matrices of two sequences Φ, Φ' on the same nodes, both of rank K, and let T be the matrix of Proposition 3.3. Then $B' = BT^\top$, $D' = (T^\top)^{-1} D$, and $P' = P$. In particular the reconstructions $\hat{f} = Pf$ coincide, and the coefficient vectors satisfy $c = T^\top c'$.*

Proof. $B'_{nk} = H'_k(x_n) = \sum_j T_{kj} H_j(x_n) = (BT^\top)_{nk}$. Then

$$D' = (TB^\top BT^\top)^{-1} TB^\top = (T^\top)^{-1}(B^\top B)^{-1} T^{-1} TB^\top = (T^\top)^{-1} D,$$

using that T is invertible (Corollary 3.4). Hence $P' = B'D' = BT^\top (T^\top)^{-1} D = BD = P$, and $c' = D'f = (T^\top)^{-1} c$. □

7 Exact reconstruction and the admissible set

Definition 7.1 (Admissible signal)**.** A function $f : [-1, 1] \to \mathbb{R}$ is *admissible of order K* if $f \in V_K$. By Proposition 3.1 this means that f is a polynomial of degree at most K with $f(0) = 0$, for every sequence Φ.

Standard fact 7.2 (Exact reconstruction on V_K)**.** Let $\operatorname{rank} B = K$ and let $p \in V_K$ with HT coordinates $a = S_K^{-1}(p)$, sampled as $f_n = p(x_n)$.

(a) The least-squares fit of the samples returns the HT of p: $Df = a$, and $\hat{f} = f$ at every node.
(b) If $K' > K$ and the basis matrix $B_{K'}$ with K' columns has rank K', then the least-squares fit of order K' of the samples is $(a, 0, \dots, 0)$: the coordinates on $H_{K+1}, \dots, H_{K'}$ are zero.

Proof. The sampled vector is $f = Ba$. By Fact 6.4, $Df = DBa = a$ and $\hat{f} = Pf = PBa = Ba = f$. For $K' > K$, $f = B_{K'}(a, 0, \dots, 0)^\top$, and the same argument with $B_{K'}$ gives the coordinates $(a, 0, \dots, 0)$. □

This is the standard fact that a left inverse reproduces the coordinates of every vector in the column space, applied to the column space (3). The content specific to the HT is the identification of the admissible set: because V_K does not depend on Φ, neither does the set of admissible signals (Proposition 3.1), while the coordinates a do depend on Φ, through T (Corollary 3.4).

Definition 7.3 (Leakage)**.** Let $f \in \mathbb{R}^N$ and let $K < K'$ with $\operatorname{rank} B_{K'} = K'$. The *leakage of f from order K to order K'* is the vector of coordinates $((D_{K'}f)_{K+1}, \dots, (D_{K'}f)_{K'})$, that is, the coordinates the order-K' least-squares fit assigns to the added basis functions $H_{K+1}, \dots, H_{K'}$. A class of signals is *leakage-free at order K* if this vector is zero for every $K' > K$ and every signal in the class.

Remark 7.4 (Leakage-free by construction)*.* Standard fact 7.2(b) states that the class V_K is leakage-free at order K. Fact 6.4 states that $f - \hat{f} = (I - P)f$ is orthogonal to $\operatorname{col}(B)$, so the decomposition $f = Pf + (I - P)f$ is exact and the error of the reconstruction is the distance from f to $\operatorname{col}(B)$; the transform contributes no other term. No periodic extension enters the construction: the basis functions are polynomials on $[-1, 1]$, nothing identifies the endpoints -1 and 1, and so the boundary mismatch whose spectrum spreads across the coefficients of a DFT expansion of a non-periodic signal does not arise. Leakage-free is a statement about the construction. For a signal outside V_K the coordinates on added basis functions are in general nonzero, and the residual is the distance from the signal to the fitted space (Proposition 7.5); that distance is approximation error and is a property of the signal and the space, not of the transform. A least-squares fit by a Fourier basis on the same nodes is also an orthogonal projection; the difference lies in the fitted space, which for a Fourier basis consists of periodic functions.

Proposition 7.5 (Signals outside V_K)**.** *(a) For every $q \in V_K$ and every function f on $[-1, 1]$, $\sup_{x \in [-1,1]} |f(x) - q(x)| \geq |f(0)|$.*
(b) A continuous function f on $[-1, 1]$ is a uniform limit of elements of $\bigcup_K V_K$ if and only if $f(0) = 0$.
(c) If 0 is a node, then the reconstruction $\hat{f}$ at that node equals 0 for every K and every Φ, so the residual there is $f(0)$.

In particular, the constant function 1, $\cos x$, and $x^2 - \frac{1}{3}$ are not admissible of any order.

Proof. (a) Every $q \in V_K$ has $q(0) = 0$, so $\sup |f - q| \geq |f(0) - q(0)| = |f(0)|$.

(b) If $q_j \to f$ uniformly with $q_j \in V_{K_j}$, then $f(0) = \lim q_j(0) = 0$. Conversely let f be continuous with $f(0) = 0$ and let $\eta > 0$. By the Weierstrass approximation theorem there is a polynomial p of degree K with $\max |f - p| < \eta/2$. Then $|p(0)| = |p(0) - f(0)| < \eta/2$, and $q = p - p(0)$ satisfies $q(0) = 0$, so $q \in V_K$ by Proposition 3.1, and $\max |f - q| \leq \max |f - p| + |p(0)| < \eta$.

(c) $\hat{f} = \sum_k c_k H_k$ at the node 0 equals $\sum_k c_k H_k(0) = 0$.

The three named functions are nonzero at 0 (values 1, 1, $-\frac{1}{3}$), so by (a) none is in any V_K. □

Proposition 7.6 (Comparison with the full polynomial space)**.** *For nodes $x_1, \dots, x_N$, $K \geq 1$, and $f \in \mathbb{R}^N$, let $e_K^V(f) = \min_{c \in \mathbb{R}^K} \|Bc - f\|_2$ and $e_K^{\mathcal{P}}(f) = \min_{p \in \mathcal{P}_K} \|(p(x_n))_n - f\|_2$. Then $e_K^{\mathcal{P}}(f) \leq e_K^V(f)$.*

Proof. By (3), $\mathrm{col}(B) \subseteq \{(p(x_n))_n : p \in \mathcal{P}_K\}$, because $V_K \subseteq \mathcal{P}_K$. A minimum over a larger set is not larger than the minimum over a subset. □

Section 13 shows that adjoining the constant function to the basis makes the two spaces equal.

8 Evaluation of the basis matrix by the recurrence

A flop is one floating-point addition, subtraction, or multiplication.

Definition 8.1 (Evaluation by the recurrence)**.** *Evaluation of the basis matrix by the recurrence* means evaluating the whole matrix B from the recurrence of Definition 1.1: all K basis functions are evaluated at every node in one pass, with $\Phi_2, \dots, \Phi_{K-1}$ computed once, and each row of B is generated by

$$B_{n1} = x_n, \qquad B_{n2} = 2x_n^2, \qquad B_{n,k+1} = 2x_n\, B_{nk} - \Phi_k\, B_{n,k-1} \quad (2 \leq k \leq K-1).$$

Evaluation by the recurrence accounts for every basis function with the same recurrence that defines the family, so B is generated from the definition itself and no expanded coefficients are required.

Proposition 8.2 (Cost of evaluation by the recurrence)**.** *For $K \geq 2$, evaluation by the recurrence forms B in $3NK - 4N$ flops, which is at most $3NK$.*

Proof. Computing $2x_n$ once per node costs N flops. Column 1 is x_n (no flops). Column 2 is $(2x_n) \cdot x_n$ (one flop per node, N in total). Each of the $K-2$ entries per node in columns $3, \dots, K$ costs one multiplication $(2x_n) \cdot B_{nk}$, one multiplication $\Phi_k \cdot B_{n,k-1}$, and one subtraction, that is 3 flops. The total is $N + N + 3N(K-2) = 3NK - 4N$. □

Proposition 8.3 (Separate evaluation of each H_k)**.** *Suppose the coefficients of $H_1, \dots, H_K$ in powers of x are known; computing them once from the recurrence of Proposition 2.1 on coefficient arrays costs $O(K^2)$ flops, independent of N. Evaluating each H_k separately at every node by Horner's rule in $y = x^2$ costs at most $N(1 + K(K+1)/2)$ flops, which is $O(NK^2)$.*

Proof. By Proposition 2.2, every power of x in H_k has the form x^{k-2j} with $0 \leq j \leq \lfloor (k-1)/2 \rfloor$, so the smallest exponent is $s_k = 1$ for odd k and $s_k = 2$ for even k, and $H_k = x^{s_k} R_k(x^2)$ with R_k a polynomial of degree $m_k = \lfloor (k-1)/2 \rfloor$. Horner's rule evaluates $R_k(y)$ with m_k multiplications and m_k additions; multiplying by x^{s_k} costs one more flop ($x \cdot R_k$ for $s_k = 1$, $y \cdot R_k$ for $s_k = 2$). So H_k costs at most $2m_k + 1 \leq k$ flops per node. Computing $y = x^2$ costs one flop per node. The total is at most $N(1 + \sum_{k=1}^{K} k) = N(1 + K(K+1)/2)$. The coefficient arrays: q_k is obtained from q_{k-1} and q_{k-2} with $O(k)$ flops, so all K of them cost $O(K^2)$. □

Propositions 8.2 and 8.3 compare two evaluation routes to the same matrix B: the recurrence costs $O(NK)$ flops and separate evaluation from expanded coefficients costs $O(NK^2)$ flops. The comparison concerns only the evaluation of the basis. It is not a statement about the cost of the least-squares fit, which is given in Section 10, and it is not a statement about the FFT.

Proposition 8.4 (Row independence)**.** *Row n of B depends only on x_n and on Φ. Let $b_n \in \mathbb{R}^{1\times K}$ denote row n. Then*

$$B^\top B = \sum_{n=1}^{N} b_n^\top b_n, \qquad B^\top f = \sum_{n=1}^{N} f_n\, b_n^\top.$$

Proof. $B_{nk} = H_k(x_n)$ depends on x_n only. $(B^\top B)_{jk} = \sum_n B_{nj}B_{nk} = \sum_n (b_n^\top b_n)_{jk}$ and $(B^\top f)_j = \sum_n B_{nj} f_n$. □

Consequently the rows can be generated independently of each other, in parallel, and $B^\top B$ and $B^\top f$ can be accumulated one node at a time with working storage of $K^2 + 2K$ numbers beyond the samples, instead of storing the NK entries of B. This route solves the explicit normal equations and is subject to the warning of Section 9.

9 Conditioning and regularization

Let B have rank K and singular values $\sigma_1 \geq \ldots \geq \sigma_K > 0$. Define $\kappa(B) = \sigma_1/\sigma_K$.

Standard fact 9.1. $\kappa(B^\top B) = \kappa(B)^2$.

Proof. With $B = U\Sigma V^\top$ the singular value decomposition, $B^\top B = V\Sigma^2 V^\top$, a symmetric matrix with eigenvalues σ_i^2. Hence its condition number is σ_1^2/σ_K^2. □

Standard fact 9.2 (Distance to singularity)**.** For a nonsingular matrix M, the smallest 2-norm of a perturbation E for which $M + E$ is singular is $\sigma_{\min}(M)$, so the relative distance $\|E\|_2/\|M\|_2$ to the nearest singular matrix is $1/\kappa(M)$ (Eckart and Young; see [3]).

The basis matrix depends on Φ, on K, and on the nodes, and so does $\kappa(B)$. Table 2 reports $\kappa(B)$ for the geometric sequence at two grid settings. At $N = 64$, $K = 15$ the smallest of the seven tabulated values occurs at $\lambda = 1$. At $N = 32$, $K = 10$ the smallest tabulated value occurs at $\lambda = 0.95$ and is below the value at $\lambda = 1$. No statement about a minimizing λ is made beyond the tabulated values.

Proposition 9.3 (Tikhonov regularization)**.** *Let* $\operatorname{rank} B = K$*, $f = Bc_0$ with $c_0 \in \mathbb{R}^K$, $\alpha > 0$, and $c_\alpha = (B^\top B + \alpha I)^{-1} B^\top f$. Then $c_\alpha - c_0 = -\alpha\,(B^\top B + \alpha I)^{-1} c_0$ and*

$$\|c_\alpha - c_0\|_2 \leq \frac{\alpha}{\sigma_K^2 + \alpha}\, \|c_0\|_2.$$

Proof. $B^\top f = B^\top B c_0 = (B^\top B + \alpha I)c_0 - \alpha c_0$, so $c_\alpha = c_0 - \alpha(B^\top B + \alpha I)^{-1} c_0$. The matrix $B^\top B + \alpha I$ is symmetric with eigenvalues $\sigma_i^2 + \alpha \geq \sigma_K^2 + \alpha$, so its inverse has 2-norm at most $1/(\sigma_K^2 + \alpha)$. □

The regularized solution is therefore not exact for admissible signals; the bound is explicit, and $\alpha \to 0$ recovers Standard fact 7.2. The parameter α is chosen by the user.

Warning for the explicit normal equations. Forming $B^\top B$ squares the condition number (Fact 9.1), and the relative distance from $B^\top B$ to the nearest singular matrix is $1/\kappa(B)^2$ (Fact 9.2). In binary64 arithmetic, with $\varepsilon = 2^{-52}$, this distance is at most ε when $\kappa(B) \geq 2^{26} = 67\,108\,864$. At that point the rounding scale of the arithmetic reaches the distance to singularity, and the explicit normal equations carry no accuracy guarantee. The threshold depends on the precision, and it is a

warning: it does not state that every computed solution is wrong. For smaller $\kappa(B)$ the error of the normal equations still grows with $\kappa(B)^2\varepsilon$. Solvers based on the QR factorization or the singular value decomposition operate on B directly [3, 7]. Section 11 specifies such a solver.

10 Cost of the least-squares fit

Proposition 10.1 (Operation count)**.** *Let $B \in \mathbb{R}^{N\times K}$ with $K \le N$. The steps of the least-squares fit computed through the normal equations (the QR and SVD routes are stated after the proof) cost:*

(i) forming B by evaluation by the recurrence: at most $3NK$ flops (Proposition 8.2);
(ii) forming $B^\top B$ using symmetry: $K(K+1)(2N-1)/2$ flops;
(iii) forming $B^\top f$: $K(2N-1)$ flops;
(iv) solving $(B^\top B)c = B^\top f$ by Cholesky factorization and two triangular solves: $K^3/3 + O(K^2)$ flops;
(v) forming $\widehat{f} = Bc$: $N(2K-1)$ flops.

The total is $O(NK^2)$.

Proof. (ii) $B^\top B$ is symmetric with $K(K+1)/2$ distinct entries; each is an inner product of length N, costing N multiplications and $N-1$ additions. (iii) K inner products of length N. (iv) Standard operation counts for Cholesky factorization and triangular substitution [3]. (v) N inner products of length K. The dominant term is the NK^2 term of (ii). Since $K \le N$, $K^3 \le NK^2$, and every other term is at most a constant times NK^2. □

A Householder QR solution of the least-squares problem costs $2NK^2 - \frac{2}{3}K^3$ flops [3], and SVD-based least-squares solvers also cost $O(NK^2)$ for $K \le N$ [3]. The solver recommended in Section 11 therefore has the same order of cost as the normal equations; it differs in numerical reliability (Section 9), not in the order of the operation count.

11 Implementation

This section lists the computational steps. It contains no new mathematics and is separate from the results above; it does not depend on a programming language or library.

Step 1. *Inputs.* Samples f_n at N distinct points of an interval $[a, b]$, an order K, and a real sequence $\Phi_2, \Phi_3, \dots$.

Step 2. *Normalize the domain.* Map each sample location $s_n \in [a, b]$ to $x_n = \dfrac{2s_n - (a+b)}{b-a} \in [-1, 1]$. This maps $a \mapsto -1$ and $b \mapsto 1$, and the origin $x = 0$ corresponds to the midpoint $s = (a+b)/2$.

Step 3. *Choose nodes when the sample locations are free.* Use $x_j = \cos((2j-1)\pi/(2N))$, $j = 1, \dots, N$, with sample locations $s_j = ((b-a)x_j + (a+b))/2$.

Step 4. *Check the rank condition.* Count the distinct nonzero nodes m and require $K \le m$ (Proposition 6.2). For Chebyshev nodes this is Corollary 6.3.

Step 5. *Assemble B by the recurrence* (Definition 8.1): column 1 is x_n; column 2 is $2x_n^2$; column $k+1$ is $2x_nB_{nk} - \Phi_k B_{n,k-1}$ for $k = 2, \dots, K-1$.

Step 6. *Compute the least-squares fit.* Solve the least-squares problem for c with a QR-based or SVD-based solver applied to B and f. If $\kappa(B) \geq 2^{26}$ in binary64, the warning of Section 9 applies and the explicit normal equations are not used; for smaller $\kappa(B)$ the error of the normal equations still grows with $\kappa(B)^2\varepsilon$.

Step 7. *Regularize if required.* If a regularized solution is wanted, solve $(B^\top B + \alpha I)c = B^\top f$ and use the bound of Proposition 9.3.

Step 8. *Reconstruct and report.* Compute $\widehat{f} = Bc$, the relative error $\|f - \widehat{f}\|_2/\|f\|_2$, and $\kappa(B)$; record the arithmetic precision, the solver, and the library versions.

The reconstruction does not depend on the choice of Φ (Corollary 6.6); the choice of Φ affects the coefficients and $\kappa(B)$. The choice $\Phi_k = 1$ requires no parameters and gives $H_k = xU_{k-1}$. Another sequence is chosen when the coefficients with respect to that sequence are the object of interest.

12 Numerical results

Results are reported in three kinds. *Exact* values are rational numbers computed in exact arithmetic with SymPy. *High-precision* values are computed with mpmath at 60 decimal digits of working precision, with the normal equations solved by LU factorization at that precision. Tables 2, 3, 4, and 5 were recomputed at 100 decimal digits, and every displayed digit agrees between the two precisions. *Binary64* values are computed in IEEE 754 double precision and depend on the library build, the processor, and the solver driver.

12.1 Exact values

Table 1 gives $H_1, \dots, H_4$ at five rational points for $\Phi_k = 1$ and for the geometric sequence with $\lambda = 6/5$, $\Phi_0 = 1$, that is $\Phi_k = (5/6)^k$ with $\Phi_1 = 5/6$, $\Phi_2 = 25/36$, $\Phi_3 = 125/216$. The polynomials are

$$H_3 = 4x^3 - \tfrac{25}{36}x, \qquad H_4 = 8x^4 - 2(\tfrac{25}{36} + \tfrac{125}{216})x^2 = 8x^4 - \tfrac{275}{108}x^2.$$

For example, $H_3(\frac{1}{2}) = 4\cdot\frac{1}{8} - \frac{25}{36}\cdot\frac{1}{2} = \frac{36}{72} - \frac{25}{72} = \frac{11}{72}$. For $\Phi_k = 1$ the entries agree with Proposition 4.2: $H_k(1) = k$ and $H_k(-1) = (-1)^k k$.

Table 1: Exact values of $H_k(x)$ at rational points.

Sequence		$x=-1$	$x=-\frac{1}{2}$	$x=0$	$x=\frac{1}{2}$	$x=1$
$\Phi_k = 1$	H_1	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1
	H_2	2	$\frac{1}{2}$	0	$\frac{1}{2}$	2
	H_3	-3	0	0	0	3
	H_4	4	$-\frac{1}{2}$	0	$-\frac{1}{2}$	4
$\Phi_k = (5/6)^k$	H_1	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1
	H_2	2	$\frac{1}{2}$	0	$\frac{1}{2}$	2
	H_3	$-\frac{119}{36}$	$-\frac{11}{72}$	0	$\frac{11}{72}$	$\frac{119}{36}$
	H_4	$\frac{589}{108}$	$-\frac{59}{432}$	0	$-\frac{59}{432}$	$\frac{589}{108}$

12.2 Condition numbers

Table 2: $\kappa(B) = \sigma_{\max}(B)/\sigma_{\min}(B)$ for $\Phi_k = \lambda^{-k}$, $\Phi_0 = 1$, on N Chebyshev nodes with K basis functions. Computed at 60 digits (identical displayed digits at 100 digits), rounded to four significant digits. $\kappa(B^\top B) = \kappa(B)^2$.

(N, K)	$\lambda = 0.9$	$\lambda = 0.95$	$\lambda = 1$	$\lambda = 1.05$	$\lambda = 1.1$	$\lambda = 1.2$	$\lambda = 2$
$(64, 15)$	3.494×10^5	976.6	103.1	6552	4.415×10^4	3.063×10^5	3.999×10^6
$(32, 10)$	323.5	30.2	43.91	227.6	562.5	1618	1.147×10^4

12.3 Reconstruction errors

The relative error is $\|f - \hat{f}\|_2 / \|f\|_2$ over the $N = 64$ Chebyshev nodes, with $\hat{f}$ the least-squares reconstruction. The computation was run for $\Phi_k = 1$ and for the geometric sequence with $\lambda = 1.1$, $\Phi_0 = 1$. Over all 36 pairs of a signal and a value of K, the two relative errors differ by at most 9.49×10^{-54} in absolute value, in agreement with Corollary 6.6. Table 3 lists the values for $\Phi_k = 1$.

Table 3: Relative error $\|f - \hat{f}\|_2 / \|f\|_2$ of the least-squares fit on $N = 64$ Chebyshev nodes, for $\Phi_k = 1$, computed at 60 digits and rounded to four significant digits. The column "in V_K" states for which K the signal is admissible.

Signal f	in V_K	$K = 5$	$K = 10$	$K = 15$	$K = 20$	$K = 25$	$K = 30$
$e^{-x}\sin(3\pi x)$	no	0.7906	0.1235	0.001293	7.55×10^{-7}	3.543×10^{-10}	8.07×10^{-14}
$\sin(\pi x)(1 - x^2)$	no	0.047	0.0001039	4.631×10^{-10}	3.106×10^{-14}	3.748×10^{-21}	4.047×10^{-26}
$x^3 - \frac{1}{2}x$	$K \geq 3$	$< 10^{-59}$	$< 10^{-59}$	$< 10^{-59}$	$< 10^{-59}$	$< 10^{-59}$	$< 10^{-59}$
1	no	0.4472	0.3015	0.2582	0.2182	0.2	0.1796
$\cos x$	no	0.5717	0.3854	0.3301	0.279	0.2557	0.2296
$x^2 - \frac{1}{3}$	no	0.3814	0.2571	0.2202	0.1861	0.1706	0.1532

The signal $x^3 - \frac{1}{2}x$ belongs to V_K for every $K \geq 3$, and its error is below 10^{-59}, that is zero within the working precision, in agreement with Standard fact 7.2. The error for $e^{-x}\sin(3\pi x)$ is 3.543×10^{-10} at $K = 25$ and 8.07×10^{-14} at $K = 30$. The signals 1, $\cos x$, and $x^2 - \frac{1}{3}$ are not admissible of any order (Proposition 7.5), and their errors remain above 0.15 at $K = 30$.

12.4 Reconstruction with the constant adjoined

Table 4 repeats the computation of Table 3 for the basis $\{1, H_1, \ldots, H_K\}$ with $\Phi_k = 1$, which spans $\mathcal{P}_K$ (Proposition 13.1). An entry $< 10^{-59}$ is zero within the working precision.

Table 4: Relative error $\|f-\widehat{f}\|_2/\|f\|_2$ of the least-squares fit with the basis $\{1, H_1, \dots, H_K\}$, $\Phi_k = 1$, on $N = 64$ Chebyshev nodes, computed at 60 digits and rounded to four significant digits.

Signal f	$K=5$	$K=10$	$K=15$	$K=20$	$K=25$	$K=30$
$e^{-x}\sin(3\pi x)$	0.7865	0.1208	0.001202	7.482×10^{-7}	3.415×10^{-10}	8.07×10^{-14}
$\sin(\pi x)(1-x^2)$	0.047	0.0001039	4.631×10^{-10}	3.106×10^{-14}	3.748×10^{-21}	4.047×10^{-26}
$x^3-\frac{1}{2}x$	$<10^{-59}$	$<10^{-59}$	$<10^{-59}$	$<10^{-59}$	$<10^{-59}$	$<10^{-59}$
1	$<10^{-59}$	$<10^{-59}$	$<10^{-59}$	$<10^{-59}$	$<10^{-59}$	$<10^{-59}$
$\cos x$	3.785×10^{-5}	9.039×10^{-13}	1.299×10^{-18}	3.793×10^{-28}	6.618×10^{-35}	1.588×10^{-45}
$x^2-\frac{1}{3}$	$<10^{-59}$	$<10^{-59}$	$<10^{-59}$	$<10^{-59}$	$<10^{-59}$	$<10^{-59}$

The signals $x^3-\frac{1}{2}x$, 1, and $x^2-\frac{1}{3}$ are polynomials of degree at most 3, so they are reconstructed exactly for every $K \geq 3$, and the signal $\cos x$ has error 3.785×10^{-5} at $K=5$ and 1.588×10^{-45} at $K=30$. Without the constant, the errors of 1, $\cos x$, and $x^2-\frac{1}{3}$ remain above 0.15 at $K=30$ (Table 3).

12.5 Leakage

Let $f(x) = x^3-\frac{1}{2}x$, which belongs to V_3. With $\Phi_k = 1$, $H_3 = 4x^3-x$ gives $x^3 = (H_3+H_1)/4$, hence $f = \frac{1}{4}H_3 - \frac{1}{4}H_1$ and the HT coordinates are $(-\frac{1}{4}, 0, \frac{1}{4})$. On the $N=64$ Chebyshev nodes the least-squares fit of order $K'=15$ returns $(-\frac{1}{4}, 0, \frac{1}{4}, 0, \dots, 0)$ with every coordinate within 10^{-59} of this vector at 60 and at 100 digits, so the leakage from order 3 to order 15 (Definition 7.3) is below 10^{-59} in every coordinate.

For comparison, Table 5 gives the fraction of the squared DFT magnitude lying outside the lowest harmonics $|j| \leq J$ for the same function sampled at the $N=64$ uniform points $u_n = -1+2(n-1)/N$, $n = 1, \dots, N$, with $c_j = \sum_n f(u_n)e^{-2\pi i\, j(n-1)/N}$ for $j = -31, \dots, 32$. The DFT is the standard Fourier pipeline for samples on a uniform grid; it treats the samples as one period.

Table 5: Leakage illustration for $f(x) = x^3-\frac{1}{2}x$. Rows 1 to 5: fraction of the squared DFT magnitude outside the lowest harmonics $|j| \leq J$, computed at 60 digits and rounded to three significant digits. Last row: largest magnitude of the least-squares coordinates on the added basis functions (Definition 7.3).

Representation	Quantity	Value
DFT of 64 uniform samples	energy outside $\lvert j\rvert \leq 1$	0.903
DFT of 64 uniform samples	energy outside $\lvert j\rvert \leq 3$	0.507
DFT of 64 uniform samples	energy outside $\lvert j\rvert \leq 8$	0.21
DFT of 64 uniform samples	energy outside $\lvert j\rvert \leq 16$	0.0895
DFT of 64 uniform samples	energy outside $\lvert j\rvert \leq 31$	0.00231
Least-squares fit of order 15 on Chebyshev nodes	coordinates on $H_4, \dots, H_{15}$	$<10^{-59}$

12.6 Orthogonality and the Jacobi matrix

The weights on $[-1,1]$, all symmetric, are $\sigma_1 = 1$, $\sigma_{\mathrm{G}} = e^{-x^2}$, $\sigma_{\mathrm{B}} = e^{-4(x-1/2)^2} + e^{-4(x+1/2)^2}$, and $\sigma_{\mathrm{C}} = \cosh(x)/\sqrt{1-x^2}$. In each case $\nu = x^2\sigma$. The first three use M-point Gauss-Legendre rules with weights multiplied by $x^2\sigma(t_i)$. The fourth uses the M-point Gauss-Chebyshev rule $t_i = \cos((2i-1)\pi/(2M))$, $\omega_i = \pi/M$, applied to $x^2\cosh x$, which accounts for the endpoint singularity

exactly. The discretized Stieltjes procedure (S1) to (S3) was run with $K = 100$ and $M = 300$ and with $M = 400$.

Exact coefficients for σ_1. Table 6 lists $\Phi_2, \dots, \Phi_9$ computed in exact rational arithmetic from the moments $\int x^{n+2}\,dx$. With these coefficients, the integrals $\int_{-1}^{1} H_j H_k\,dx$ for $1 \le j < k \le 8$ are zero in exact arithmetic, in agreement with Proposition 5.2. Table 6 is computed in exact arithmetic, and Tables 7 to 9 are computed by the discretized Stieltjes procedure. For σ_1 and $k = 2, \dots, 9$, the 60-digit Stieltjes values differ from the exact values of Table 6 by at most 3.11×10^{-61}, and the 100-digit values by at most 1.43×10^{-100}. Exact values are not available for the other three weights.

Table 6: Exact recurrence coefficients Φ_k for $\sigma_1 = 1$ on $[-1, 1]$ ($\nu = x^2\,dx$), computed in exact rational arithmetic; the decimal column is rounded to eight places.

k	Φ_k	decimal
2	$\frac{12}{5}$	2.40000000
3	$\frac{16}{35}$	0.45714286
4	$\frac{100}{63}$	1.58730159
5	$\frac{64}{99}$	0.64646465
6	$\frac{196}{143}$	1.37062937
7	$\frac{48}{65}$	0.73846154
8	$\frac{108}{85}$	1.27058824
9	$\frac{256}{323}$	0.79256966

Coefficients for the four weights. Table 7 lists Φ_2, Φ_3, Φ_4 and Φ_{99}, Φ_{100}, Φ_{101}. The first coefficients depend on the weight. For $96 \le k \le 101$ every coefficient of every weight satisfies $|\Phi_k - 1| \le 0.0212$, and the values alternate on the two sides of 1 with the parity of k.

Table 7: Recurrence coefficients Φ_k of $\nu = x^2\sigma$ from the discretized Stieltjes procedure with $K = 100$, $M = 400$, computed at 60 digits and rounded to seven places. The 100-digit values agree in every displayed digit.

Weight	Φ_2	Φ_3	Φ_4	Φ_{99}	Φ_{100}	Φ_{101}
σ_1	2.4000000	0.4571429	1.5873016	0.9799250	1.0201255	0.9803201
σ_{G}	2.1168007	0.5453345	1.5246972	0.9800258	1.0200253	0.9804169
σ_{B}	2.1003671	0.4945494	1.6145850	0.9798374	1.0202139	0.9802359
σ_{C}	3.0987897	0.2883124	1.7210697	0.9797504	1.0202496	0.9801523

Quadrature, precision, and orthogonality. Table 8 reports, for each weight, the largest difference between the coefficients $\Phi_2, \dots, \Phi_{101}$ computed with $M = 300$ and with $M = 400$ at 60 and at 100 digits, the largest difference between the binary64 and the 60-digit coefficients, and the largest shift $|a_k|$. A difference at the level of the working precision means that the two rules agree to working precision. Table 9 reports the check of orthogonality and of the Gauss rule at $K = 30$: the largest normalized off-diagonal entry $|G_{jk}|/\sqrt{G_{jj}G_{kk}}$ of the Gram matrix $G_{jk} = \int H_j H_k\,d\sigma$

for $1 \le j \ne k \le 30$, computed with the $M = 400$ rule for σ while the coefficients come from the $M = 300$ rule, and the error of the 30-point Gauss rule of Standard fact 5.3, built from the Jacobi matrix with $\mu_0 = \sum_i w_i$, for $\cos 3x$, e^x, $1/(1+x^2)$, and the relative error for x^{58}, which has degree $2K - 2 = 58$. Reference integrals use the $M = 400$ rule for ν. At 60 digits the Gram entries and the errors for $\cos 3x$, e^x, and x^{58} are below 10^{-58}, and at 100 digits the Gram entries and the errors for e^x and x^{58} are below 10^{-98}, which is the working precision in each case; the error for $\cos 3x$ at 100 digits is the truncation error of the rule, since $\cos 3x$ is not a polynomial and the 30-point rule is exact only up to degree 59: the Chebyshev coefficient of degree 60 of $\cos 3x$ is $2J_{60}(3) = 8.517 \times 10^{-72}$, the scale of the observed errors. The error for $1/(1+x^2)$ is identical at both precisions and is also the truncation error of the rule: the poles at $\pm i$ have Bernstein ellipse parameter $1 + \sqrt{2}$, and $(1+\sqrt{2})^{-60} = 1.08 \times 10^{-23}$ is the scale of the observed errors.

Table 8: Checks of the Stieltjes computation. Δ_M: largest difference between the coefficients $\Phi_2, \dots, \Phi_{101}$ from the $M = 300$ and the $M = 400$ rules, at 60 and at 100 digits. Δ_{64}: largest difference between the binary64 and the 60-digit coefficients. $\max|a_k|$: largest shift at 60 digits.

Weight	Δ_M (60)	Δ_M (100)	Δ_{64}	$\max\|a_k\|$
σ_1	1.4×10^{-60}	8.57×10^{-101}	1.33×10^{-15}	0
σ_{G}	1.4×10^{-60}	9.29×10^{-101}	9.99×10^{-16}	0
σ_{B}	1.01×10^{-60}	1.0×10^{-100}	9.99×10^{-16}	0
σ_{C}	2.64×10^{-60}	2.0×10^{-100}	1.11×10^{-15}	1.08×10^{-60}

Table 9: Orthogonality and Gauss rule at $K = 30$. Gram: largest normalized off-diagonal entry at 60 and at 100 digits. The remaining columns, at 100 digits: absolute error of the 30-point Gauss rule of Standard fact 5.3 for $\cos 3x$, e^x, $1/(1+x^2)$, and relative error for x^{58}.

Weight	Gram (60)	Gram (100)	$\cos 3x$	e^x	$1/(1+x^2)$	x^{58}
σ_1	2.23×10^{-60}	2.47×10^{-100}	3.32×10^{-72}	2.86×10^{-101}	4.8×10^{-24}	1.63×10^{-100}
σ_{G}	1.75×10^{-60}	2.37×10^{-100}	2.01×10^{-72}	4.29×10^{-101}	3.17×10^{-24}	8.96×10^{-101}
σ_{B}	2.12×10^{-60}	2.6×10^{-100}	2.25×10^{-72}	7.86×10^{-101}	3.48×10^{-24}	2.08×10^{-100}
σ_{C}	7.32×10^{-61}	1.09×10^{-100}	8.36×10^{-72}	1.43×10^{-100}	9.93×10^{-24}	1.26×10^{-99}

A weight that is not symmetric. For $\nu = x^2 e^x$ on $[-1, 1]$ the shifts are $a_0 = 0.511452$, $a_1 = -0.381874$, $a_2 = 0.237016$, $a_3 = -0.193712$, computed at 60 and at 100 digits with identical displayed digits. These are nonzero, so the orthogonal polynomials of this weight are not produced by the recurrence of Definition 1.1.

12.7 The rank condition

Table 10 tests Proposition 6.2 on Chebyshev nodes for odd and even N. The proposition gives $\operatorname{rank} B = K$ exactly when $m \ge K$. In all seven cases the smallest singular value is of order 1 when $m \ge K$, and its displayed digits are unchanged at 100 digits. When $m < K$ it is zero in exact arithmetic; the computed values 9.517×10^{-62} and 1.109×10^{-61} at 60 digits become 2.391×10^{-102} and 2.785×10^{-102} at 100 digits, tracking the working precision.

Table 10: Test of Proposition 6.2 on Chebyshev nodes. m is the number of distinct nonzero nodes; "$m \geq K$" is the condition of the proposition; $\sigma_{\min}(B)$ is computed at 60 digits.

N	K	m	$m \geq K$	$\sigma_{\min}(B)$
5	4	4	yes	0.6549
5	5	4	no	9.517×10^{-62}
6	5	6	yes	0.4641
6	6	6	yes	0.4641
7	6	6	yes	0.5013
7	7	6	no	1.109×10^{-61}
8	8	8	yes	0.3978

12.8 Binary64 observations

Table 11 reports relative errors computed in binary64 for $N = 64$, with $\kappa(B)$ from the high-precision computation. "SVD" is the SVD-based least-squares solver; "NE" solves the explicit normal equations. The nodes are computed in binary64. The exact error of $x^3 - \frac{1}{2}x$ is below 10^{-59} (Table 3), so the binary64 entries for that signal are rounding error. At $\lambda = 1.1$, $K = 25$, $\kappa(B) = 7.589 \times 10^8 > 2^{26}$, and the NE errors exceed the SVD errors by factors between 88 and 139 for the three signals.

Table 11: Binary64 relative errors on $N = 64$ Chebyshev nodes (environment in Section 12, Reproducibility). $\kappa(B)$ is the 60-digit value. These values depend on the computing environment.

λ	K	$\kappa(B)$	Solver	$e^{-x}\sin(3\pi x)$	$\sin(\pi x)(1-x^2)$	$x^3 - \frac{1}{2}x$
1	15	103.1	SVD	1.29×10^{-3}	4.63×10^{-10}	5.71×10^{-15}
			NE	1.29×10^{-3}	4.63×10^{-10}	6.80×10^{-15}
1	25	273.3	SVD	3.54×10^{-10}	5.02×10^{-15}	1.64×10^{-14}
			NE	3.54×10^{-10}	3.00×10^{-14}	1.30×10^{-14}
1.1	15	4.415×10^4	SVD	1.29×10^{-3}	4.63×10^{-10}	1.18×10^{-14}
			NE	1.29×10^{-3}	4.63×10^{-10}	4.42×10^{-13}
1.1	25	7.589×10^8	SVD	5.38×10^{-10}	3.92×10^{-11}	2.13×10^{-11}
			NE	7.47×10^{-8}	3.92×10^{-9}	1.89×10^{-9}

12.9 Reproducibility

The grid is the N Chebyshev nodes $x_j = \cos((2j-1)\pi/(2N))$ with $N = 64$ unless stated (Table 2 also uses $N = 32$, and Table 10 uses the N listed there). The sequence is $\Phi_k = \lambda^{-k}$ ($\Phi_0 = 1$). The adjoined-constant basis uses the columns $1, H_1, \ldots, H_K$ with $\Phi_k = 1$. The signals are $e^{-x}\sin(3\pi x)$, $\sin(\pi x)(1-x^2)$, $x^3 - \frac{1}{2}x$, 1, $\cos x$, and $x^2 - \frac{1}{3}$, evaluated at the nodes. The error norm is $\|f - \hat{f}\|_2 / \|f\|_2$ over the nodes. The high-precision values use mpmath 1.3.0 at 60 and 100 decimal digits, with $\kappa(B)$ computed as the square root of the ratio of the extreme eigenvalues of $B^\top B$. The binary64 values were computed with Python 3.12.3, NumPy 2.4.4 with OpenBLAS (scipy-openblas build) 0.3.31.188.0, on x86_64, using `numpy.linalg.lstsq` with its default arguments for SVD and `numpy.linalg.solve` on $B^\top B$ for NE. Digits of the binary64 results beyond the leading one or two can differ in another environment.

13 Scope and domain of applicability

Domain. The recurrence basis is defined on $[-1, 1]$; a signal on $[a, b]$ is mapped by Step 2 of Section 11, and the origin of the normalized domain is the midpoint $(a + b)/2$.

Representable signals. The signals represented exactly by the HT of order K are the elements of V_K (Standard fact 7.2), which are the polynomials of degree at most K that vanish at the midpoint of the domain. A continuous signal is a uniform limit of such polynomials if and only if it vanishes at the midpoint (Proposition 7.5). A signal that does not vanish at the midpoint is outside this set, and no reconstruction from the recurrence basis reproduces the value of the signal there.

Relation to the full polynomial space. For fixed K and fixed nodes, the least-squares error of the fit in the recurrence basis is at least that of the degree-K polynomial fit (Proposition 7.6); for signals in V_K both errors are zero. The restriction to V_K is removed by adjoining the constant function:

Proposition 13.1 (Adjoining the constant)**.** *Let Φ be any real sequence.*

(a) The functions $1, H_1, \dots, H_K$ span $\mathcal{P}_K$.

(b) Let B^+ be the matrix with $K + 1$ columns formed by the column $\mathbf{1}$ followed by the columns of B. Then $\operatorname{rank} B^+ = K + 1$ if and only if the nodes contain at least $K + 1$ distinct values.

(c) For such nodes the least-squares reconstruction from B^+ equals the least-squares degree-K polynomial fit on the nodes, hence equals the least-squares fit by Chebyshev or Legendre polynomials of degree K on the same nodes, and every polynomial of degree at most K is reconstructed exactly.

Proof. (a) $V_K \subseteq \mathcal{P}_K$ and $1 \in \mathcal{P}_K$. A constant c with $c(0) = 0$ is zero, so $V_K \cap \operatorname{span}\{1\} = \{0\}$, and $\dim(\operatorname{span}\{1\} + V_K) = 1 + K = \dim \mathcal{P}_K$.

(b) Suppose $B^+c = 0$ and let $p = c_0 + \sum_k c_k H_k \in \mathcal{P}_K$; then p vanishes at every node. If there are at least $K+1$ distinct nodes, p has at least $K+1$ roots and degree at most K, so $p = 0$, and $c = 0$ because $\{1, H_1, \dots, H_K\}$ is a basis of $\mathcal{P}_K$ by (a). If there are $d \le K$ distinct nodes $z_1, \dots, z_d$, then $p = \prod_i (x - z_i) \in \mathcal{P}_K$ is nonzero and vanishes at every node, so it equals $c_0 + \sum_k c_k H_k$ for some $c \neq 0$ with $B^+c = 0$.

(c) $\operatorname{col}(B^+) = \{(p(x_n))_n : p \in \mathcal{P}_K\}$ by (a), and the orthogonal projection onto a subspace does not depend on the basis chosen for it (Fact 6.4). Exactness for $\mathcal{P}_K$ follows as in Standard fact 7.2. □

With the constant adjoined, the admissible set is all polynomials of degree at most K, the value of the signal at the midpoint is no longer constrained, and the comparison of Proposition 7.6 becomes an equality of spaces. The recurrence basis as defined in Section 1 keeps $H_0 = 0$; the adjoined variant is a different space, $\mathcal{P}_K$ instead of V_K.

Other bases. On $[-1, 1]$, Chebyshev and Legendre expansions are the standard tools for non-periodic signals, and on Chebyshev nodes the Chebyshev coefficients are computed in $O(N \log N)$ operations by a discrete cosine transform [10, 1, 8], while the least-squares fit costs $O(NK^2)$ (Proposition 10.1). For periodic signals the Fourier basis is the standard representation; the recurrence basis consists of polynomials on $[-1, 1]$ and contains no periodicity.

When to use which. Table 12 states which representation is the standard choice in each situation. It is a table of tool selection and contains no claim that one representation is better than another outside its situation.

Table 12: Standard choice of representation by situation.

Situation	Representation	Reason
Periodic signal sampled on a uniform grid over whole periods	DFT or FFT	$O(N \log N)$; the periodicity of the signal matches the periodicity of the basis.
Non-periodic signal on $[a, b]$ reconstructed by a polynomial least-squares fit, no periodic extension	Least-squares fit in the recurrence basis	Orthogonal projection onto V_K; exact on V_K with zero coordinates on added basis functions (Standard fact 7.2).
Arbitrary distinct sample locations	Least-squares fit in the recurrence basis	The fit needs at least K distinct nonzero nodes (Proposition 6.2); the standard FFT needs a uniform grid.
Smooth non-periodic signal, the full degree-K polynomial space wanted	Chebyshev or Legendre expansion, or the least-squares fit with the constant adjoined	Same space $\mathcal{P}_K$ (Proposition 13.1); Chebyshev coefficients on Chebyshev nodes by a DCT in $O(N \log N)$.
Large N or large K	Fast transform methods suited to the grid and signal: FFT for periodic uniform samples, DCT for Chebyshev-grid polynomial fits	The least-squares fit costs $O(NK^2)$ (Proposition 10.1) and $\kappa(B)$ grows with K (Table 11).
Signal with $f(0) = 0$ known from the model or the construction, moderate N and K	Least-squares fit in the recurrence basis with $\Phi_k = 1$	Parameter-free basis; exact on V_K.

Coefficients. The coefficient vector depends on Φ through the unit lower triangular map of Corollary 3.4 and Corollary 6.6; the reconstruction does not.

Weights. For a symmetric finite positive measure σ on $[-1, 1]$ with infinite support, Section 5 gives a recurrence basis orthogonal for σ and the Gauss rule of $x^2\sigma$, and Section 12 reports the computed cases. A non-symmetric weight needs a shift term that the recurrence of Definition 1.1 does not contain; the least-squares fit applies to such data without change.

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